

LABWORK 4

This lab aims to help you practice loops and basic algorithms by solving **real-life inspired problems** using **functions** in C. You will implement 10 functions, each focusing on a different type of algorithmic thinking — calculation, iteration, and pattern generation.

Function 1: sum_first_n_numbers (5 pts)

Return Type: `int` **Parameters:** `int n` **Description:** Calculate the sum of the numbers from 1 to n.

Example: Input: 5 Output: 15 (1+2+3+4+5)

Function 2: print_number_square (5 pts)

Return Type: `void` **Parameters:** `int n` **Description:** Print a square of numbers from 1 to n, where each row is filled with the same numbers.

Example: Input: 4 Output: 1 1 1 1 2 2 2 2

Function 3: print_number_sequence (5 pts)

Return Type: `void`

Parameters: `int n`

Description: Prints numbers from 1 to n in a single line separated by spaces.

Example:

Input: 5 Output: 1 2 3 4 5

Function 4: print_multiples_of_five

Return Type: `void` **Parameters:** `int n` **Description:** Print the multiples of 5 from 1 to n.

Example: Input: 20 Output: 5 10 15 20... n

Function 5: print_reverse_numbers

Return Type: `void` **Parameters:** `int n` **Description:** Print the numbers from 1 to n in reverse order.

Example: Input: 5 Output: 5 4 3 2 1

Function 6: print_rainfall_pattern (10 pts)

Return Type: `void`

Parameters: `int days`

Description: Prints a pattern of rainfall using `*`, where each line represents a day and the number of stars decreases by 1 (simulating decreasing rainfall).

Example:

For `days = 4`:

```
Day 1: *****
Day 2: ****
Day 3: ***
Day 4: **
```

Function 7: `print_water_intake_table` (10 pts)

Return Type: `void`

Parameters: `int glasses`

Description: Prints the total water intake in liters for 1 to the given number of glasses.

Assume 1 glass = 0.25 liters.

Example:

For `glasses = 5`:

```
1 glass = 0.25 liters
2 glasses = 0.5 liters
3 glasses = 0.75 liters
4 glasses = 1.0 liters
5 glasses = 1.25 liters
```

Function 8: `print_hollow_square` (10 pts)

Return Type: `void` **Parameters:** `int size` **Description:** Print a square of size `size`, where the outer edges are `*` (stars), and the inner is filled with space characters.

Example: For `size = 5`:

```
*****
*  *
*  *
*  *
*  *
*****
```

Function 9: `print_cinema_seat_pattern` (10 pts)

Return Type: `void`

Parameters: `int rows`

Description: Prints the seating layout of a cinema hall.

Each row has 5 seats labeled `[A] [B] [C] [D] [E]`.

Example:

For `rows = 3`:

```
[A] [B] [C] [D] [E]
[A] [B] [C] [D] [E]
[A] [B] [C] [D] [E]
```

****Function 10: print_bus_schedule**

Return Type: `void` **Parameters:** `int stops` **Description:** Print the bus schedule for the given number of stops, where each stop is 5 minutes apart.

Example: Input: 3 stops Output: Stop 1: 5 minutes Stop 2: 10 minutes Stop 3: 15 minutes ... Stop n: (n-1)*5 minutes

Main Function (15 pts)

Function Name: `main`

Return Type: `int`

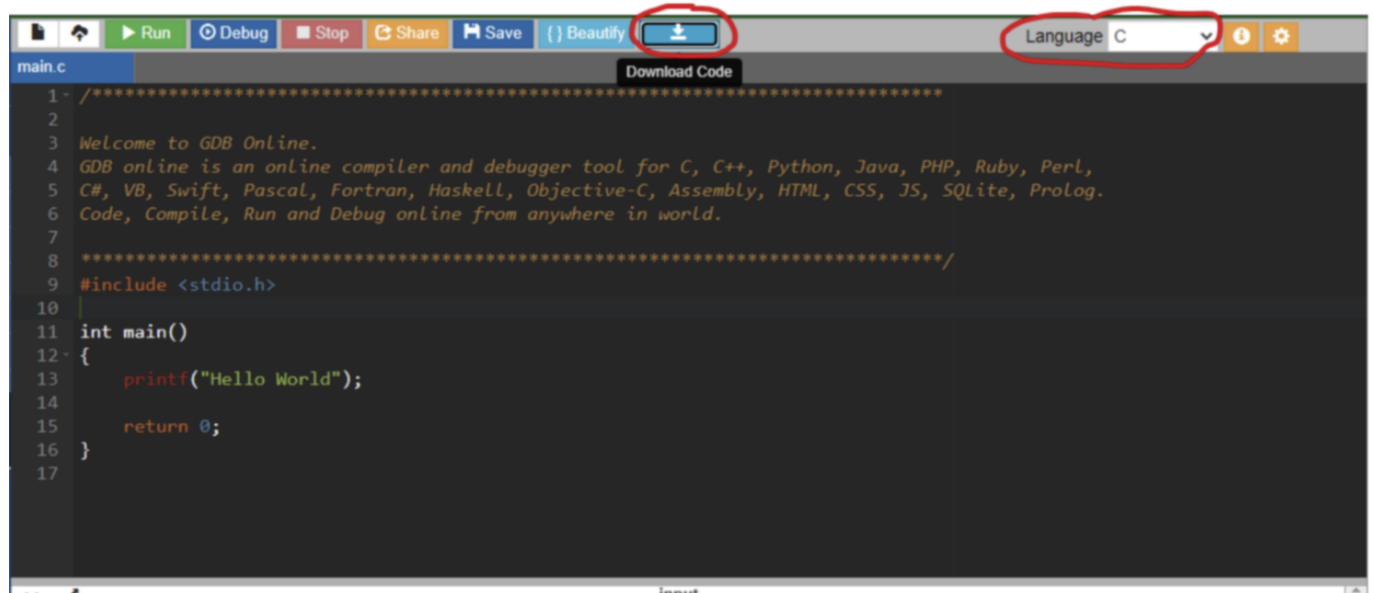
Parameters: none

Description:

- Ask the user for input arrays and numbers.
 - Call each of the 10 functions.
 - Print the results clearly and formatted.
 - Demonstrate proper input validation and loop usage.
-

Submission Format

- Submit your `.c` file named as:
`name_surname.c`
- Late submissions will not be accepted.
- Follow function naming and signature exactly.
- Clear screen and print statements are optional, not graded.



```
1  /******  
2  
3  Welcome to GDB OnLine.  
4  GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,  
5  C#, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, Prolog.  
6  Code, Compile, Run and Debug online from anywhere in world.  
7  
8  *****/  
9  #include <stdio.h>  
10  
11 int main()  
12 {  
13     printf("Hello World");  
14  
15     return 0;  
16 }  
17
```

Important Notes

- You are encouraged to modularize your code by creating functions where applicable.
- If you're unsure about the use of specific features or need assistance, refer to the course material or ask your instructor.